

PROJECT SCOPE STATEMENT	
Project Name	Ecosystem Discovery Application
Project Deliverables	Detailed Description
Freshwater Ecosystem Interface	A web-based interactive ecosystem scene representing a single Saskatchewan-inspired freshwater ecosystem. The interface will display 6–8 clickable animals positioned within ecologically appropriate zones such as sky, shoreline, water surface, and underwater areas. Users will be able to explore the environment visually and interact with animals through simple click-based interactions. The design will emphasize child-friendly usability, including: • Large clickable areas • Bright, readable visuals • Simple and intuitive navigation suitable for children ages 7–9
Animal Information Cards	When a user selects an animal, the system will display a pop-up information panel containing: • Animal name • Cartoon-style animal illustration • Short educational description written at Grade 2–3 reading level • Basic ecological information such as habitat or behavior These cards provide the educational learning component of the application.
Sticker-Book Collection Page	The application will include a digital sticker-book panel that tracks discovery progress. Features include: • Discovered animals appear as unlocked stickers • Undiscovered animals appear as locked placeholders • A progress indicator (e.g., <i>3 of 6 animals discovered</i>) • Immediate updates after an animal is discovered This feature reinforces learning through reward-based interaction.
Structured Data Model	The system will include a structured data model storing: • Animal information (name, description, habitat zone, image reference) • Discovery status (locked/unlocked) • User progress information Data will be stored using structured JSON data or lightweight local storage and updated as users interact with the ecosystem.

MVC Architecture implementation	The application will follow the Model–View–Controller (MVC) architectural pattern: Model • Stores animal data and discovery status View • Displays the ecosystem scene, sticker book, and information cards Controller • Handles user interactions such as clicking animals, unlocking stickers, and updating progress This separation ensures the system is organized, maintainable, and scalable.
Functional MVP Application	The project will deliver a working Minimum Viable Product (MVP) demonstrating the complete discovery interaction cycle: Ecosystem exploration → Animal selection → Information display → Sticker unlock → Progress update → Data persistence. The MVP will include: • At least 6 animals • One example of negative ecosystem impact (e.g., pollution) • One example of positive environmental restoration interaction (e.g., garbage cleanup)
Project Documentation	The project deliverables will include formal documentation: • Stakeholder Register • Project Requirements Document • Updated Scope Statement • UML-style diagrams (process, data flow, class) • MVP plan and task breakdown
Version Control Repository	The project will maintain a structured GitHub repository containing: • All source code • Organized documentation folders • Project README • GitHub Project Board (Kanban) for task tracking
Project Exclusions	
The following items are explicitly outside the scope of this project: • Multiplayer functionality • User accounts or authentication systems • Advanced physics simulations or 3D game engines • Multiple ecosystems or habitat environments • More than 8 animals in the ecosystem • Complex scoring or competitive game mechanics	

